Name : Sanika Mangrulkar

PRN : 202501050015

Practical 4

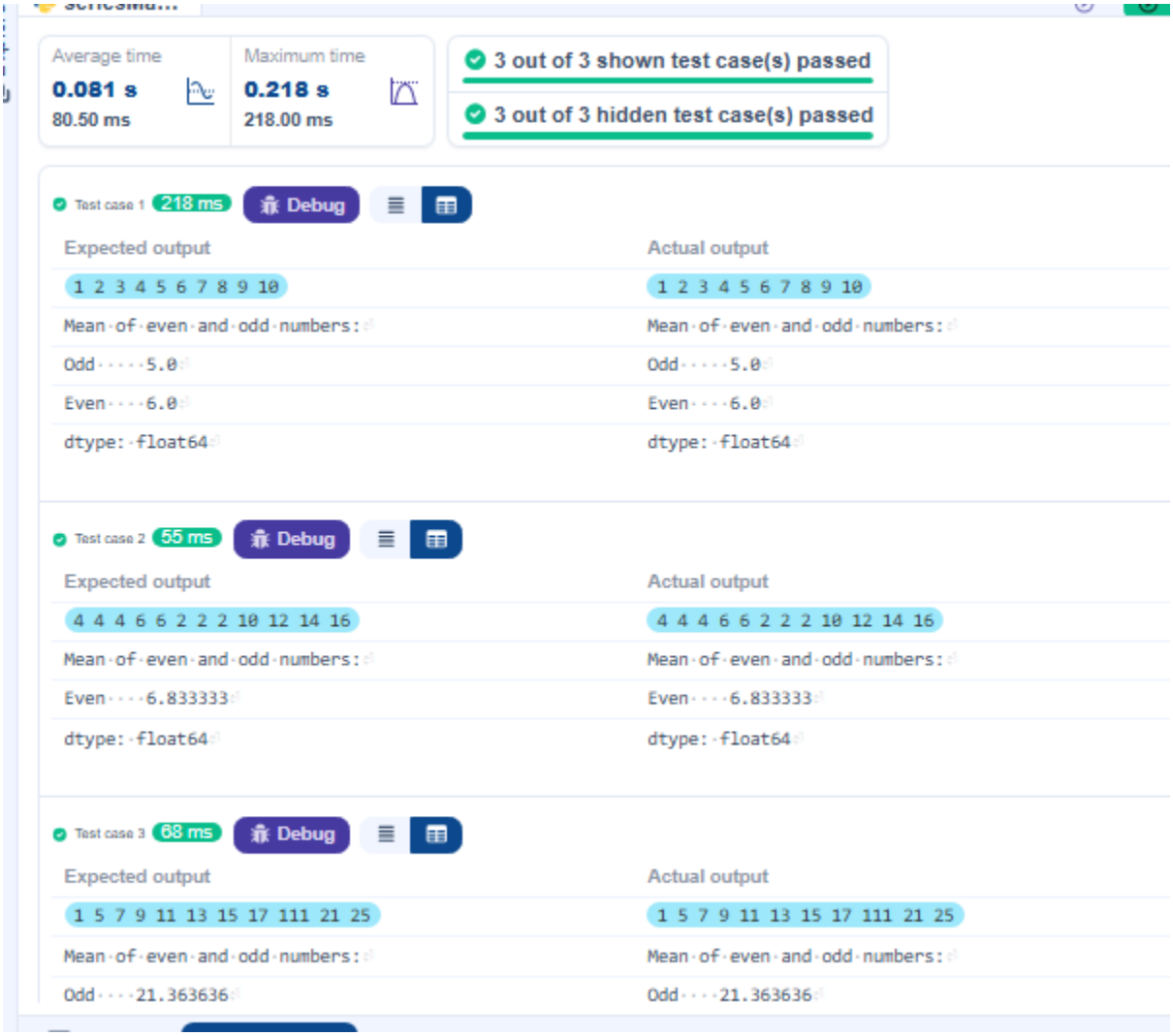
---

1)

Program :

```
import pandas as pd

# Take inputs from the user to create a list of numbers

numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers

series = pd.Series(numbers)

# Grouping by even and odd numbers and calculating the mean

grouped = series.groupby(series % 2 == 0).mean()

# Display the mean of even and odd numbers with labels

grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]

print("Mean of even and odd numbers:")

print(grouped)
```

Output :



2)

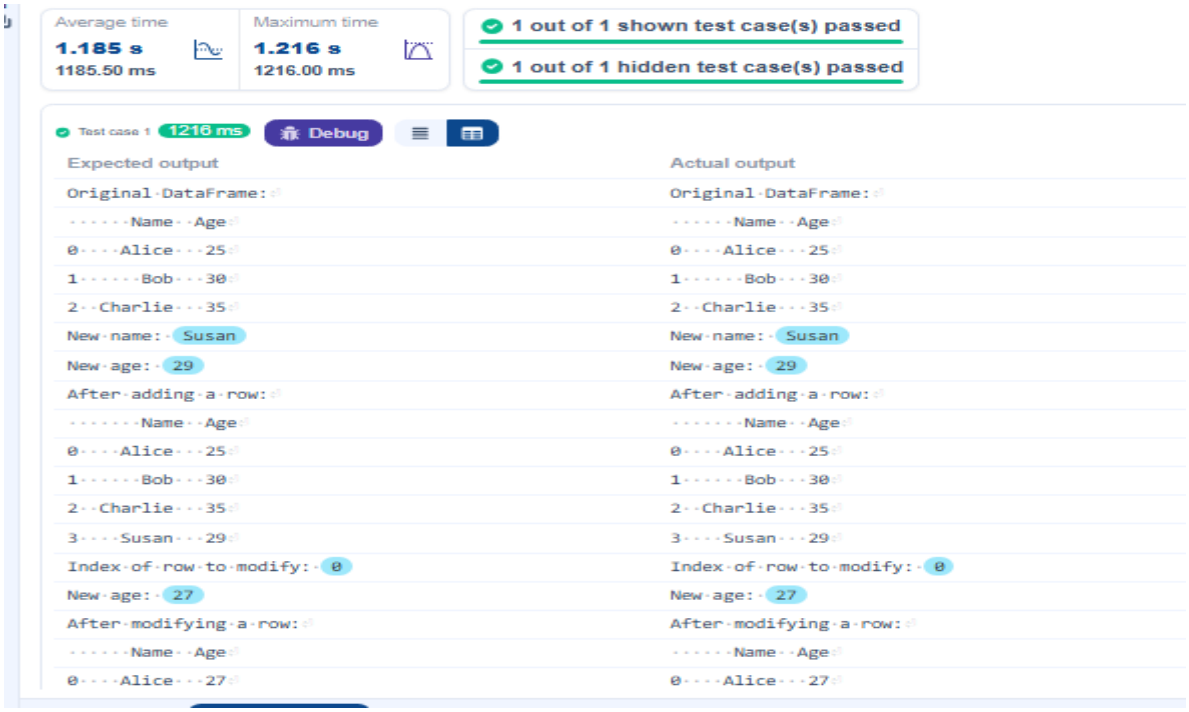
Program :

```
import pandas as pd


# Provided dictionary of lists
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
}
```

```
# Convert the dictionary to a DataFrame
df = pd.DataFrame(data)

# Display the original DataFrame
print("Original DataFrame:")
print(df)

# Adding a new row
new_name=input("New name: ")
new_age=int(input("New age: "))
new_row={'Name':new_name,'Age':new_age}
df=df.append(new_row,ignore_index=True)

# Display the DataFrame after adding a new row
print("After adding a row:\n",df)

# Modifying a row
modify_row_index=int(input("Index of row to modify: "))
new_ag_value=int(input("New age: "))
df.at[modify_row_index,'Age']=new_ag_value


# Display the DataFrame after modifying a row
print("After modifying a row:")
print(df)

# Deleting a row
```

```
delete_row_index=int(input("Index of row to delete: "))
df=df.drop(delete_row_index).reset_index(drop=True)
# Display the DataFrame after deleting a row
print("After deleting a row:")
print(df)

# Adding a new column
genders=input("Enter genders separated by space: ").split()
df['Gender']=genders

# Display the DataFrame after adding a new column
print("After adding a new column:")
print(df)

# Modifying a column
df['Name']=df['Name'].str.upper()
# Display the DataFrame after modifying a column
print("After modifying a column:")
print(df)

# Deleting a column
df=df.drop(columns=['Age'])
# Display the DataFrame after deleting a column
print("After deleting a column:")
print(df)
```

Output :



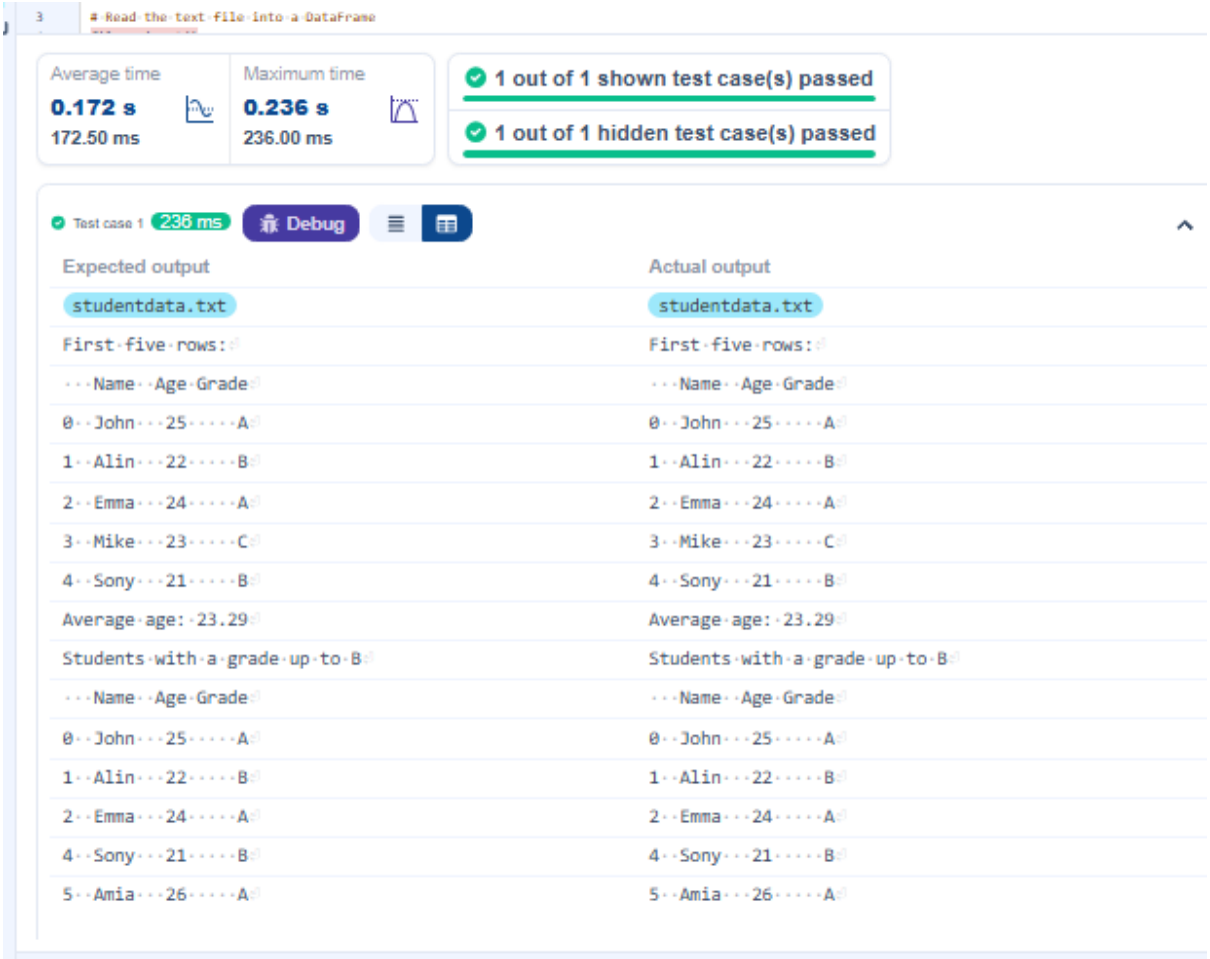
3)

Program :

```
import pandas as pd


# Read the text file into a DataFrame

file = input()

data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])


# Display the first five rows

print("First five rows:")

print(data.head())


# Calculate the average age of sstudents(limited to 2 decimal places)

average_age = round(data["Age"].mean(),2)
```

```
print(f"Average age: {average_age}")


# Filter students with a grade up to 'B'

filtered_students = data[data["Grade"] <= "B"]

print("Students with a grade up to B")

print(filtered_students)
```

Output :



4)

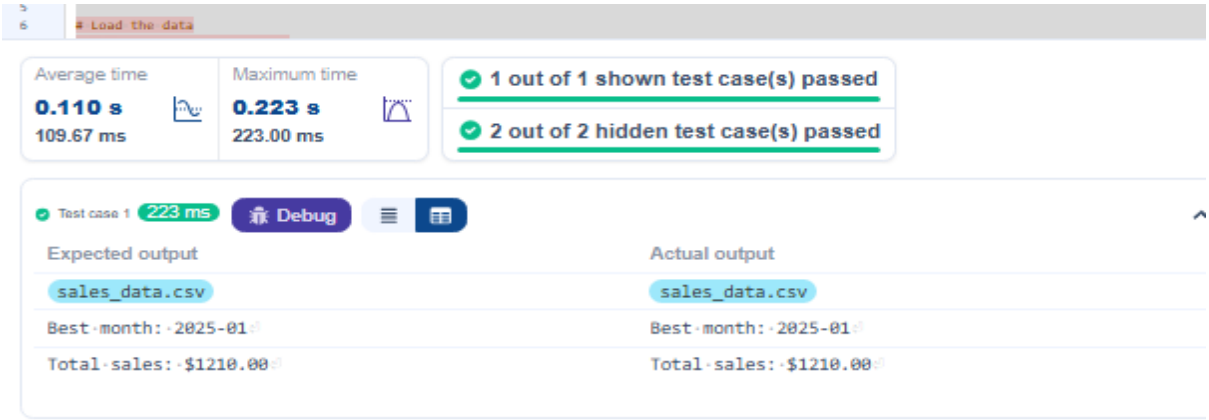
Program :

```
import pandas as pd


# Prompt the user for the file name

file_name = input()


# Load the data

df = pd.read_csv(file_name)


# Convert the Date column to datetime format

df['Date'] = pd.to_datetime(df['Date'])


# Add a Year-Month column by formatting the Date column

df['YearMonth'] = df['Date'].dt.to_period('M')


# Calculate total sales for each row

df['Total_Sales'] = df['Quantity'] * df['Price']


# Group the data by Year-Month and calculate total sales for each month

monthly_sales = df.groupby('YearMonth')['Total_Sales'].sum()


# Find the month with the highest total sales

best_month = monthly_sales.idxmax()

highest_sales = monthly_sales.max()
```

print(f"Best month: {best_month}")

print(f"Total sales: ${highest_sales:.2f}")

Output :



---

5)

Program :

import pandas as pd

Prompt the user for the file name

file_name = input()

Load the data

df = pd.read_csv(file_name)

product_sales = df.groupby('Product')['Quantity'].sum()

Find the product with the highest total quantity sold

best_product = product_sales.idxmax()

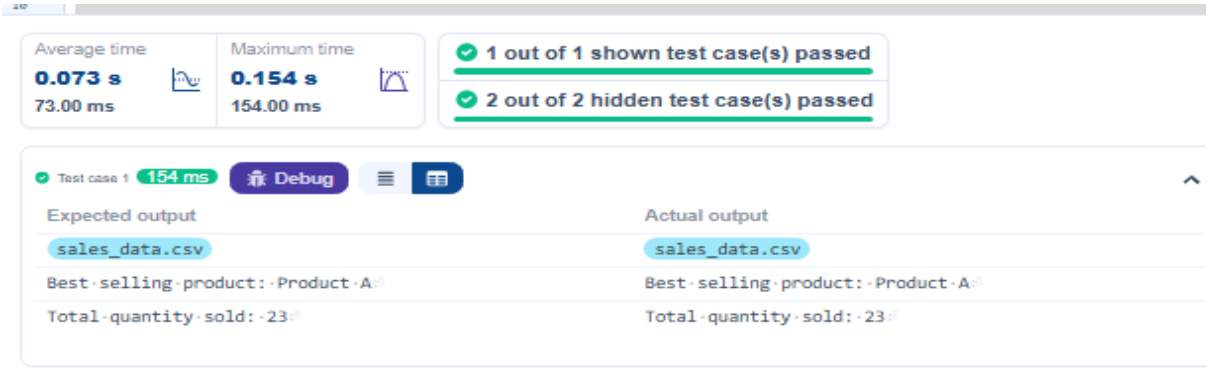
highest_quantity = product_sales.max()

Display the result

print(f"Best selling product: {best_product}")

print(f"Total quantity sold: {highest_quantity}")

Output :



---

6)

Program :

import pandas as pd

Prompt the user for the file name

```
file_name = input()


# Load the data

df = pd.read_csv(file_name)


city_sales = df.groupby('City')['Quantity'].sum()


# Find the city that sold the most products

best_city = city_sales.idxmax()

highest_quantity = city_sales.max()

# write the code..



# Display the result

print(f"City sold the most products: {best_city}")
```

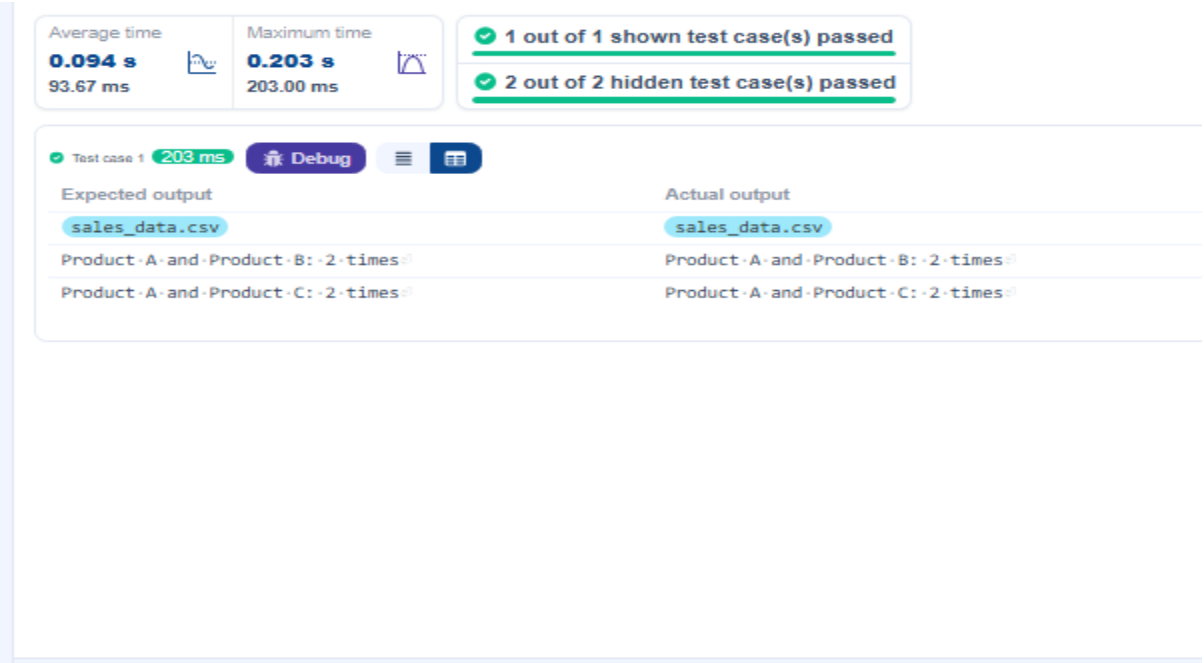
Output :

7)

Program :

```
import pandas as pd
from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)

count_pairs=Counter()
# write the code

for date,group in df.groupby("Date"):
    products = group['Product']
    for pair in combinations(sorted(products), 2):
        count_pairs[pair] +=1

# Output the most frequent product pairs

best_count= max(count_pairs.values())
best_pair= [pair for pair,val in count_pairs.items() if val==best_count ]
for pair in best_pair:
    print(f"{pair[0]} and {pair[1]}: {best_count} times")
```

Output :



8)

Program :

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset

print(data.head(5))


# 2. Display the last 5 rows of the dataset

print(data.tail(5))


# 3. Get the shape of the dataset

print(data.shape)
```

4. Get a summary of the dataset (info)

print(data.info())

5. Get basic statistics of the dataset

print(data.describe())

6. Check for missing values

print(data.isnull().sum())

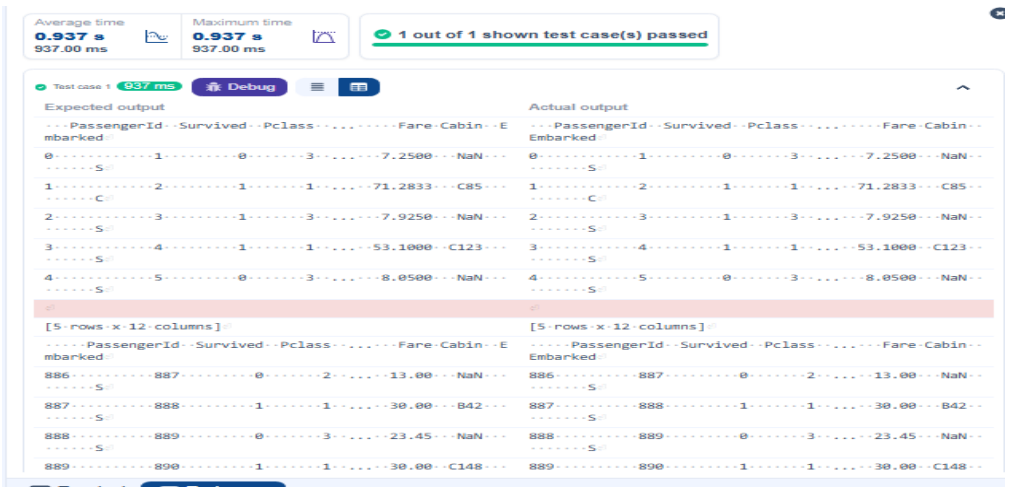
7. Fill missing values in the ‘Age’ column with the median age

8. Fill missing values in the ‘Embarked’ column with the mode

9. Drop the ‘Cabin’ column due to many missing values

10. Create a new column 'FamilySize’ by adding ‘SibSp' and ‘Parch'

Output :



9)

Program :

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']

df = pd.DataFrame(data)

# 1. Create a new column ‘IsAlone' (1 if alone, 0 otherwise)

df['FamilySize'] = df['SibSp'] + df['Parch']

df['IsAlone'] = (df['FamilySize'] == 0).astype(int)

# 2. Convert ‘Sex' to numeric (male: 0, female: 1)

df['Sex'] = df['Sex'].map({'male' : 0, 'female' : 1})


# 3. One-hot encode the ‘Embarked' column

df = pd.get_dummies(df, columns=['Embarked'])


# 4. Get the mean age of passengers

mean_age = df['Age'].mean()




# 5. Get the median fare of passengers

median_fare = df['Fare'].median()




# 6. Get the number of passengers by class
```

```
passengers_by_class = df['Pclass'].value_counts()



# 7. Get the number of passengers by gender

passengers_by_gender = df['Sex'].value_counts()



# 8. Get the number of passengers by survival status

passengers_by_survival = df['Survived'].value_counts()



# 9. Calculate the survival rate

survival_rate = df['Survived'].mean()



# 10. Calculate the survival rate by gender

survival_rate_by_gender = df.groupby('Sex')['Survived'].mean()

print(mean_age)

print(median_fare)

print(passengers_by_class)

print(passengers_by_gender)

print(passengers_by_survival)

print(survival_rate)

print(survival_rate_by_gender)
```

Output :

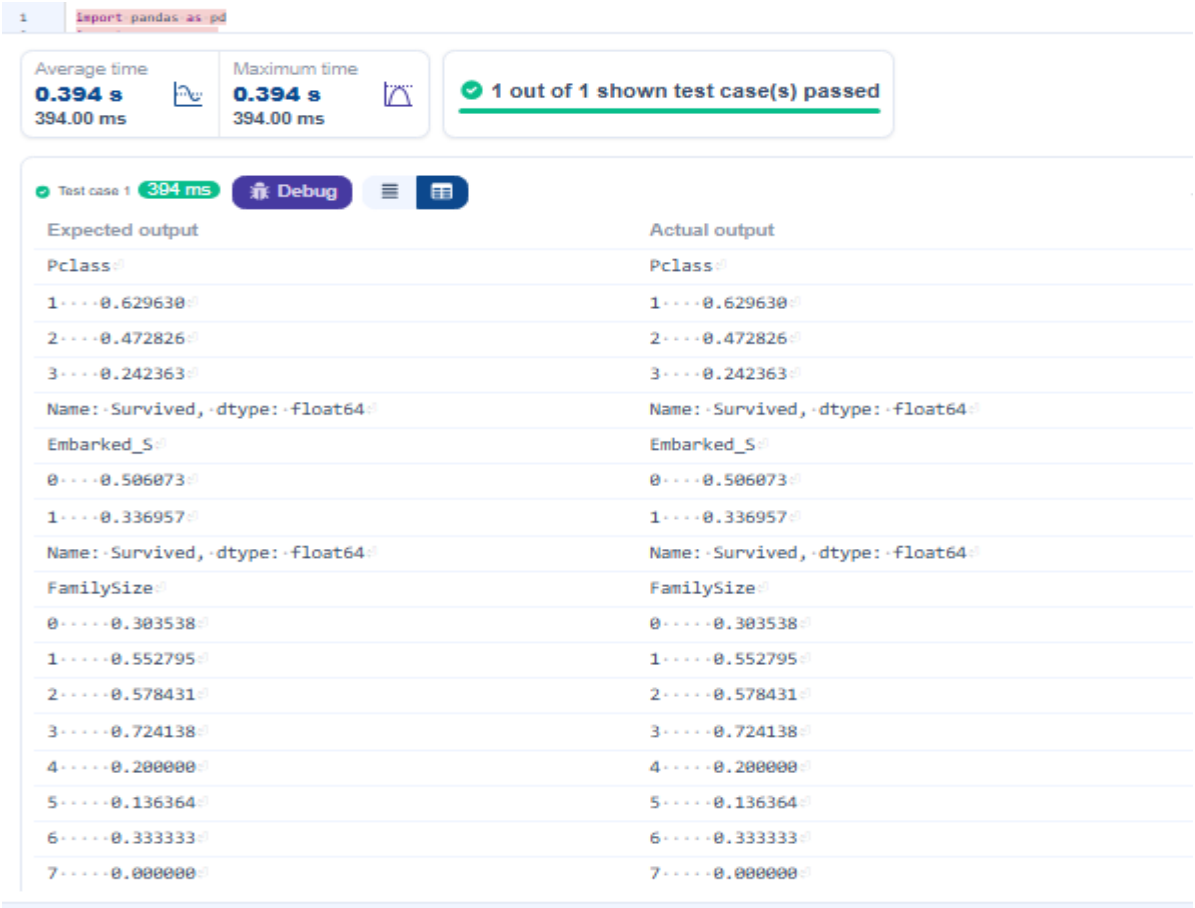
10)

Program :

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']

data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

df = pd.DataFrame(data)

# 1. Calculate the survival rate by class

grouped = df.groupby('Pclass')['Survived'].mean()

print(grouped)

# 2. Calculate the survival rate by embarked location

print(df.groupby('Embarked_S')['Survived'].mean())
```

```
# 3. Calculate the survival rate by family size
print(df.groupby('FamilySize')['Survived'].mean())

# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())

# 5. Get the average fare by class
print(data.groupby('Pclass')['Fare'].mean())

# 6. Get the average age by class
print(data.groupby('Pclass')['Age'].mean())

# 7. Get the average age by survival status
print(data.groupby('Survived')['Age'].mean())

# 8. Get the average fare by survival status
print(data.groupby('Survived')['Fare'].mean())

# 9. Get the number of survivors by class
print(data[data['Survived']==1]['Pclass'].value_counts())

# 10. Get the number of non-survivors by class
print(data[data['Survived']==0]['Pclass'].value_counts())
```

Output :



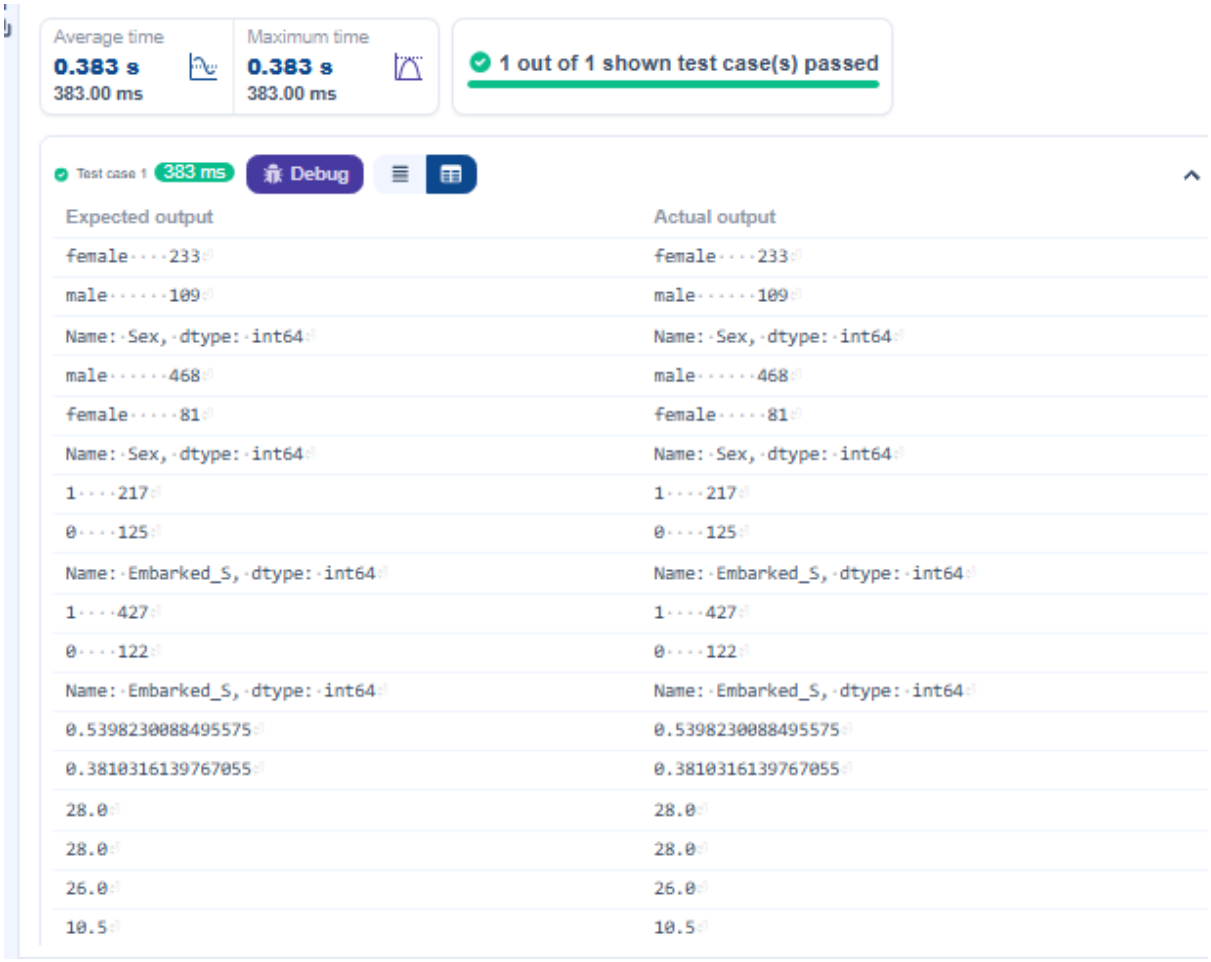
11)

Program :

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

df = pd.DataFrame(data)

# 1. Get the number of survivors by gender

print(df[df['Survived']==1]['Sex'].value_counts())

# print((df.groupby('Sex')['Survived']==1).value_counts())
```

```
# 2. Get the number of non-survivors by gender
print(data[data['Survived']==0]['Sex'].value_counts())

# 3. Get the number of survivors by embarked location
print(data[data['Survived']==1]['Embarked_S'].value_counts())

# 4. Get the number of non-survivors by embarked location
print(data[data['Survived']==0]['Embarked_S'].value_counts())

# 5. Calculate the percentage of children (Age < 18) who survived
print(data[data['Age']<18]['Survived'].mean())

# 6. Calculate the percentage of adults (Age >= 18) who survived
print(data[data['Age']>17]['Survived'].mean())

# 7. Get the median age of survivors
print(data[data['Survived']==1]['Age'].median())

# 8. Get the median age of non-survivors
print(data[data['Survived']==0]['Age'].median())

# 9. Get the median fare of survivors
print(data[data['Survived']==1]['Fare'].median())

# 10. Get the median fare of non-survivors
print(data[data['Survived']==0]['Fare'].median())
```

Output :



Expected output	Actual output
female····233	female····233
male·····109	male·····109
Name:·Sex,·dtype:·int64	Name:·Sex,·dtype:·int64
male·····468	male·····468
female····81	female····81
Name:·Sex,·dtype:·int64	Name:·Sex,·dtype:·int64
1····217	1····217
0····125	0····125
Name:·Embarked_S,·dtype:·int64	Name:·Embarked_S,·dtype:·int64
1····427	1····427
0····122	0····122
Name:·Embarked_S,·dtype:·int64	Name:·Embarked_S,·dtype:·int64
0.5398230088495575	0.5398230088495575
0.3810316139767055	0.3810316139767055
28.0	28.0
28.0	28.0
26.0	26.0
10.5	10.5